

Generated Question Paper

- Q1.** (a) Describe how the structure of a neurone relates to its function. [3 marks]
- Q2.** Scientists had higher breathing rates in the 550 ppm CO₂ _____ 2nd plot. [2 marks]
- Q3.** The a) The possible causes of increased atmospheric carbon dioxide. [3 marks]
- Q4.** (ii) Name the process that causes gas bubbles to form in the dough. [1 marks]
- Q5.** (ii) Enzymes are the chemical elements found in all proteins. [2 marks]
- Q6.** The advantage of using a pyramid of energy is that it is more efficient. [2 marks]
- Q7.** _____ (ii) State a conclusion for the results shown in Table [1 marks]
- Q8.** (b) (i) Name the source of energy used by organism A. [1 marks]
- Q9.** Enzymes are catalysts. (a) (i) Describe what is meant by the term catalyst. [2 marks]
- Q10.** The average photosynthesis rate of soybean plants was measured in two plots with different CO₂ _____ 2nd concentrations, 370 ppm and 550 ppm. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

The atmospheric carbon dioxide concentrations in the two plots were kept at:

- 370 ppm, which is similar to the current atmospheric carbon dioxide concentration
- 550 ppm, which is a possible future atmospheric carbon dioxide concentration.

When the soybean plants were fully grown, scientists calculated the average rates of photosynthesis at regular intervals from 04:00 to 22:00 for both plots.

The results are shown in Fig. 2.1.

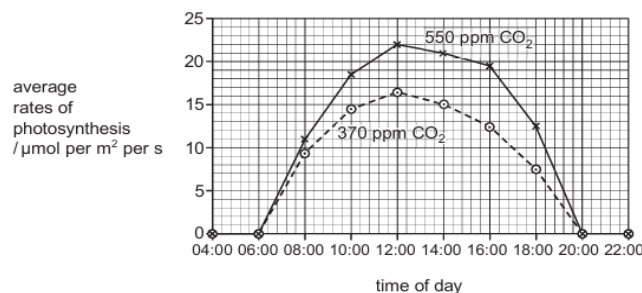


Fig. 2.1

Figure: photosynthesis_rate_graph.png

Q11. Plants store starch in their leaves. [2 marks]

Q12. The temperature of the atoms in the nucleus of the atoms is the same as the temperature of the nucleus of the atom. [2 marks]

Q13. (e) shows a synapse and parts of two neurons. (e) begins center (i) Draw an arrow to show the direction of the signal across the synapse. [1 marks]

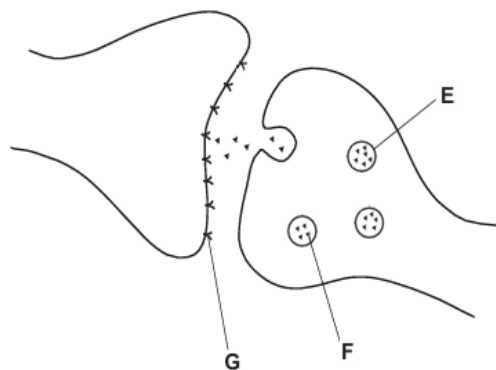


Fig. 4.1

Figure: synapse_diagram.png

Q14. (b) Lugworms live in sand on beaches and feed on diatoms, which require ammonium ions. Researchers tested effects of microplastics on lugworms; see Table 3.1. Describe and explain why researchers concluded that non-biodegradable microplastics are most harmful to lugworms using Table 3.1. [4 marks]

Q15. The effect of pH on lipase activity is shown on Fig.1. [6 marks]

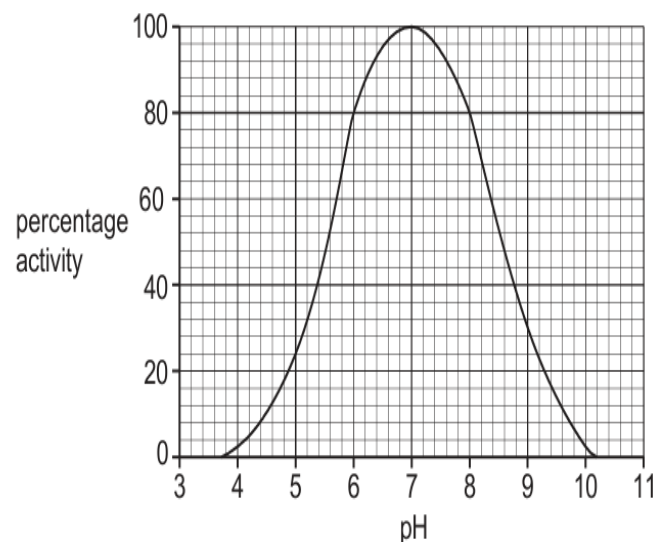


Figure: Fig1_1_LipaseActivity.png

Q16. The retina is a part of the retina. What is the retina's function? [3 marks]

Q17. (f) Discuss the effects of non-biodegradable plastics on terrestrial ecosystems. [5 marks]

Q18. Plants growing in Europe begintabularcc toprule Plant Number & Leaf Area (cm\$2\$) midrule 1 & 310 2 & 335 3 & 390 4 & 348 5 & 365 midrule Mean & 350 bottomrule endtabular (i) Using the data in Table 5.2, calculate the mean leaf area for plants growing in North America. Give your answer as a whole number and include the unit. [2 marks]

Q19. (a) State the hormone released from the testes. [1 marks]

Q20. The testes are a part of the human body. Meiosis is the process of dividing the testes into two parts. [3 marks]

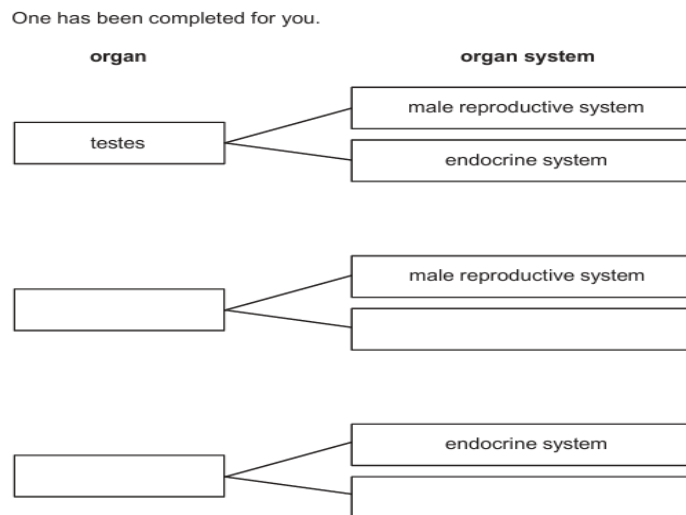


Fig. 5.1

Figure: organs_two_systems.png

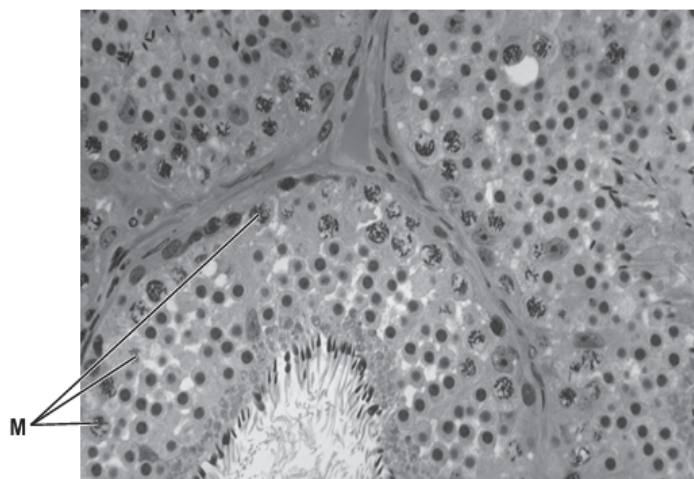


Fig. 5.2

Figure: mammalian_testis_micrograph.png

Q21. The right ventricle is located at the center of the heart. [1 marks]

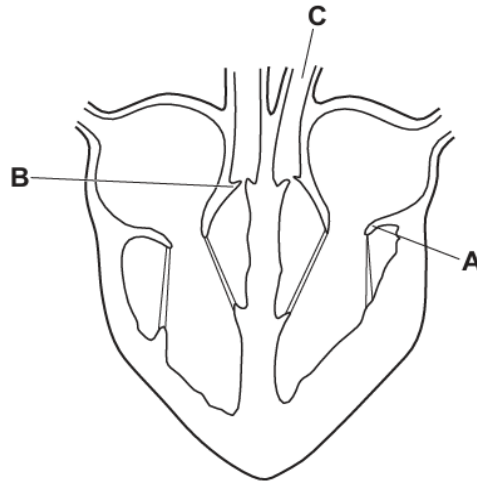


Fig. 2.1

Figure: Fig2_1_HeartCrossSection.png

Q22. A sperm cell is a cell that divides into two individual cells. A sperm cell is a cell that divides into two individual cells. A sperm cell is a cell that divides into two individual cells. [3 marks]



Fig. 5.3

Figure: sperm_cell_micrograph.png

Q23. (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q24. The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. [2 marks]

Q25. The name of organism A is a comma. [1 marks]

Biotechnology is used in the process of bread-making.

Fig. 1.1 shows some of the steps in making bread.

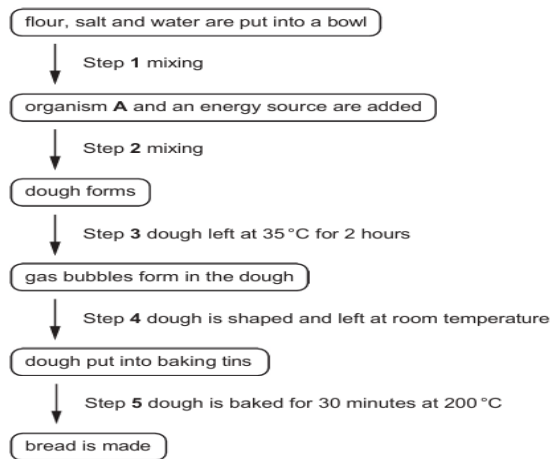


Fig. 1.1

Figure: bread_making_steps.png